Welcome

Day 1

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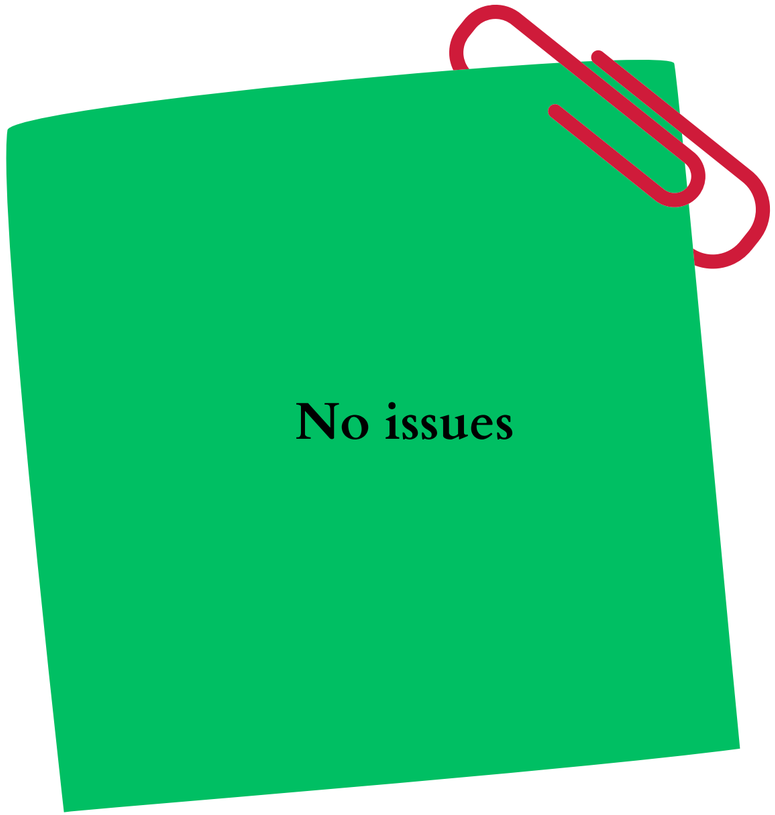
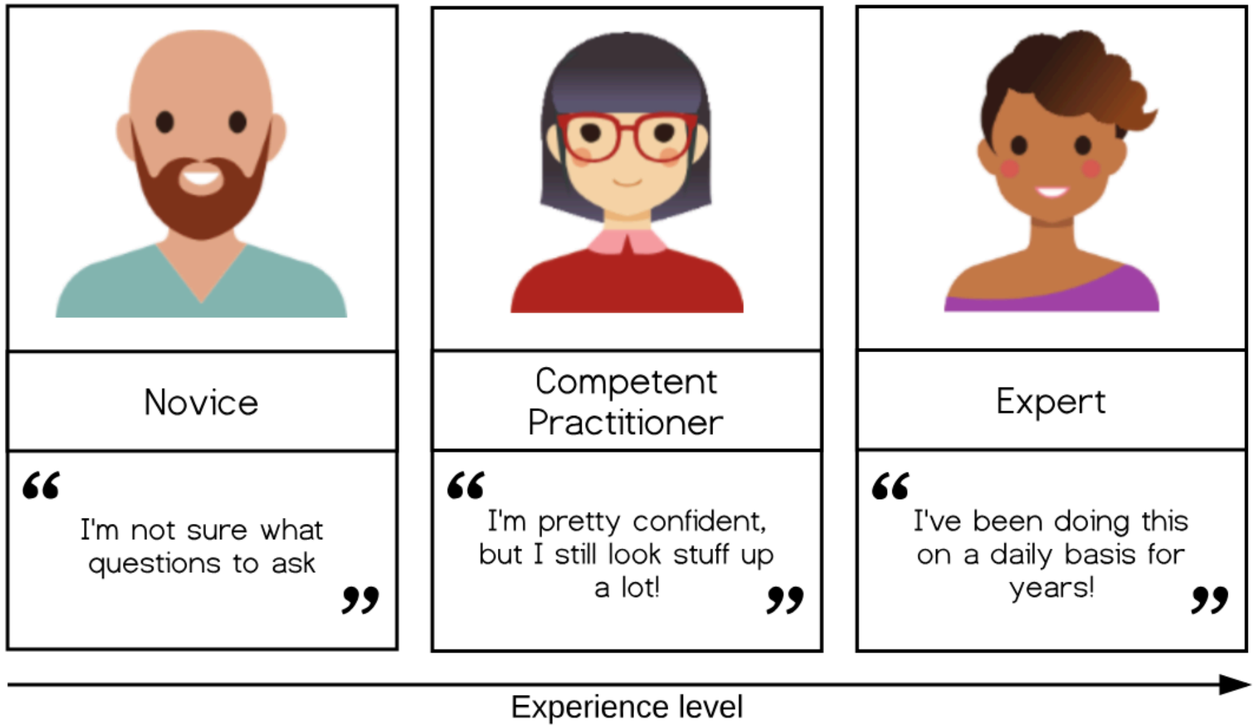
Logistics



Next break will be at noon for Lunch!

Goals


Novice
“I'm not sure what questions to ask”
Competent Practitioner
“I'm pretty confident, but I still look stuff up a lot!”
Expert
“I've been doing this on a daily basis for years!”
Experience level

YOUR FEEDBACK IS IMPORTANT

What questions do you have?

Overview

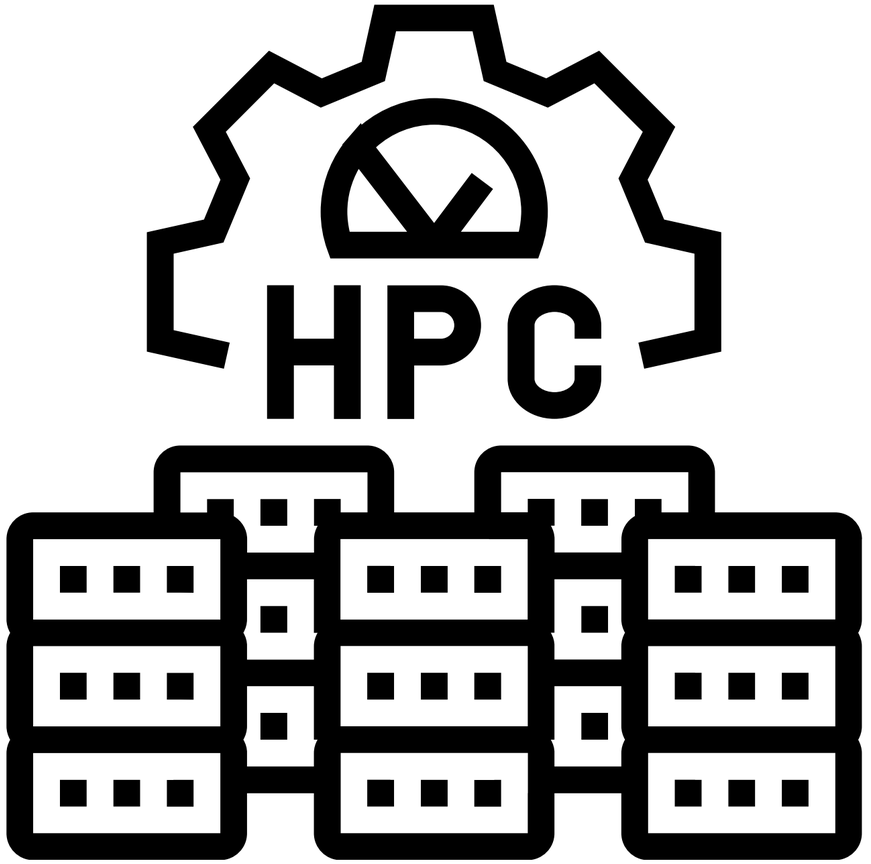
- Conceptual overview of resources
- Help & Policies
- Connecting to UAHPC
- Running your first job on UAHPC
- SFTP
- Modules and package managers

UAHPC is a set of high-performance servers...

...connected by a very high-speed network

...with storage attached to the network

...and resources that can be scheduled.



When to utilize HPC

HPC focuses on any computing that challenges the resources at hand

When your tasks are taking too long

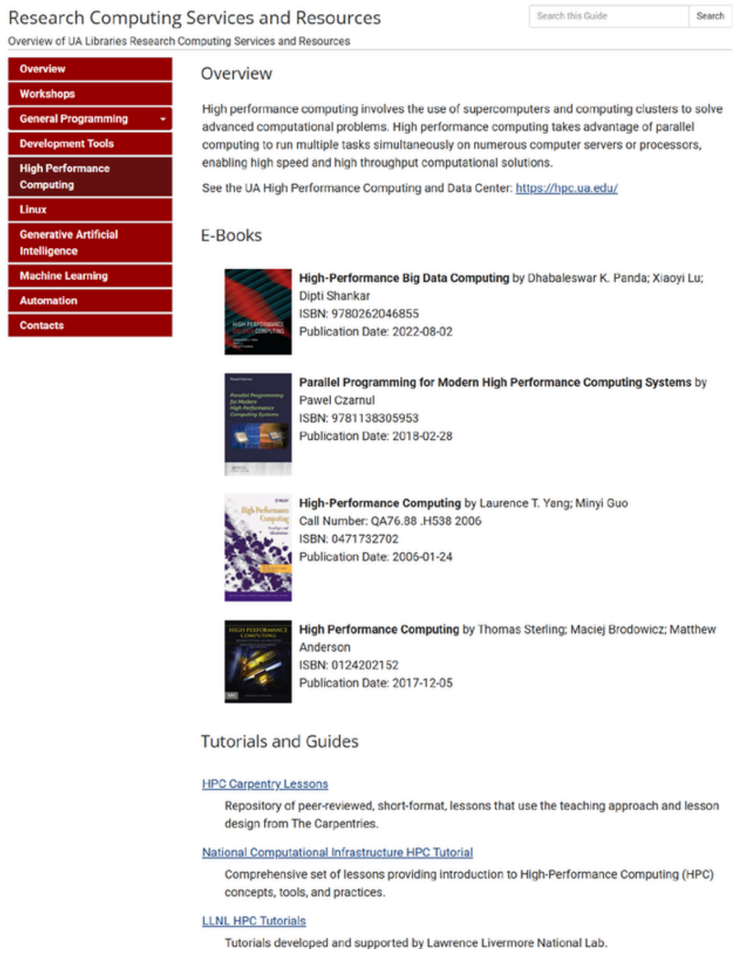
When large tasks need to be divided into smaller tasks and executed simultaneously (Parallel Computing)

When a task requires rapid data transfer between computing nodes to reduce bottlenecks

How to get help

1. Email ITSD: itsd@ua.edu
2. HPC Portal (may be faster)
3. https://guides.lib.ua.edu/RCS/hpc
4. Book an appt with me:

   https://ua.libcal.com/appointments/dubose



Key policies to know

- UAHPC is accessible from the University of Alabama IP address space.
- From off-campus, you must use the VPN to get to the UA network. The system provides services via SSH/SFTP.
- Responsible use is required.

Acceptable Use Policy | High Performance Computing and Data Center

Information Protection Procedure | Office of Information Technology

Connecting to a server

To connect to a Linux server, you need to SSH (Secure Shell) into the server.

SSH is a network protocol that allows you to securely connect to a remote server over an unsecured network.

IP name or hostname of server

Username

Password (or SSH key)

To SSH into a server, open a terminal or Powershell prompt and type the following command:

```
ssh username@hostname
```

Connecting to UAHPC

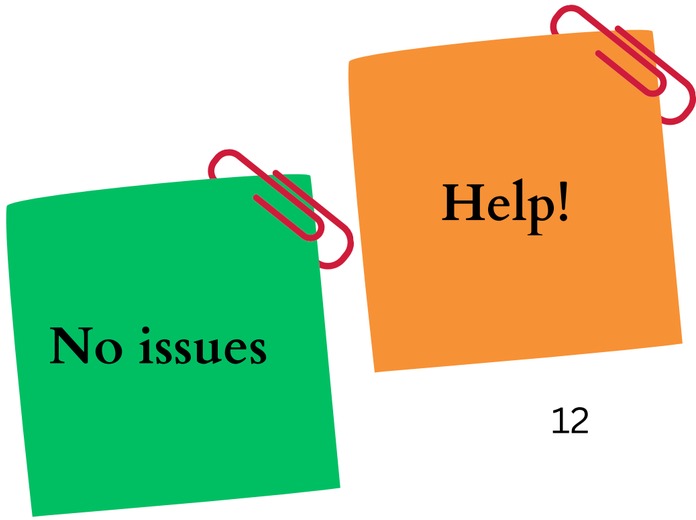
1) Open your terminal on Linux and Mac; on Windows, use Powershell
2) Enter the following command to ssh into the HPC cluster:

```
ssh myBamaID@uahpc.ua.edu
```

username

hostname

3) Enter your myBama password
4) Complete Duo two-factor authentication



Anatomy of a Linux command

Command: Built-in instruction you want to run.

Option: Also called "flags". These modify the behavior of the command and start with either a - or --.

Argument: This is the target that the command acts upon. It can be a filename, path, input, etc.

Basic Commands

- **pwd** - Print working directory or where you are currently located in the file system
- **ls** - List the contents of the current directory
- **cd** - Change directory
- **mkdir** - Create a new directory
- **rm** - Remove a file
- **rmdir** - Remove a directory
- **cp** - Copy a file
- **mv** - Move a file
- **cat** - Display the contents of a file

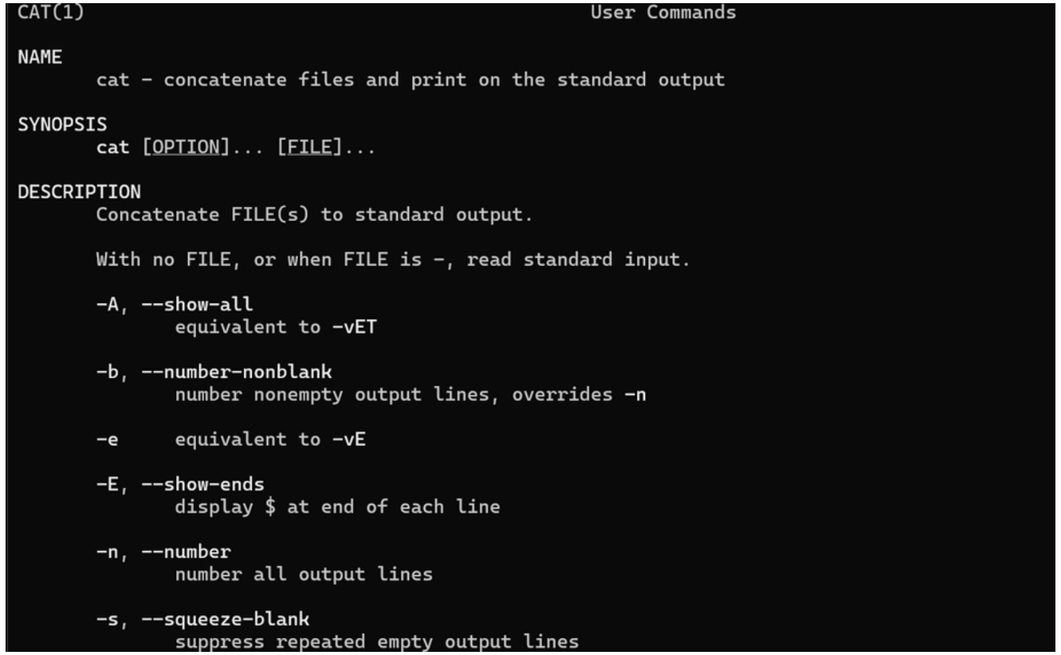
How to get help

help - Get help on a command

```
[myBamaID@uahpc-login001 ~]$ help -d cd
cd - Change the shell working directory
```

man - Display the manual page for a command

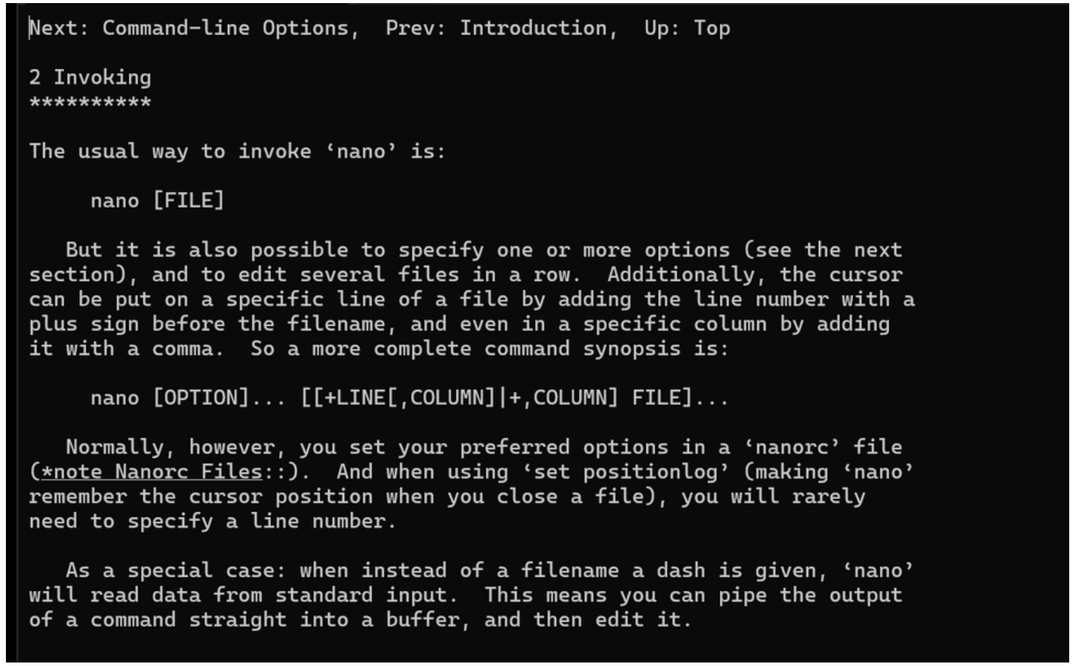
```
[myBamaID@uahpc-login001 ~]$ man cat
```



```
CAT(1)                                          User Commands

NAME
       cat - concatenate files and print on the standard output

SYNOPSIS
       cat [OPTION]... [FILE]...

DESCRIPTION
       Concatenate FILE(s) to standard output.

       With no FILE, or when FILE is -, read standard input.

       -A, --show-all
              equivalent to -vET

       -b, --number-nonblank
              number nonempty output lines, overrides -n

       -e     equivalent to -vE

       -E, --show-ends
              display $ at end of each line

       -n, --number
              number all output lines

       -s, --squeeze-blank
              suppress repeated empty output lines
```

info - offers more detailed and structured info than man command

```
[myBamaID@uahpc-login001 ~]$ info --show-options nano
```



```
Next: Command-line Options,  Prev: Introduction,  Up: Top

2 Invoking
**********

The usual way to invoke ‘nano’ is:

     nano [FILE]

   But it is also possible to specify one or more options (see the next
section), and to edit several files in a row.  Additionally, the cursor
can be put on a specific line of a file by adding the line number with a
plus sign before the filename, and even in a specific column by adding
it with a comma.  So a more complete command synopsis is:

     nano [OPTION]... [[+LINE[,COLUMN]|+,COLUMN] FILE]...

   Normally, however, you set your preferred options in a ‘nanorc’ file
(*note Nanorc Files::).  And when using ‘set positionlog’ (making ‘nano’
remember the cursor position when you close a file), you will rarely
need to specify a line number.

   As a special case: when instead of a filename a dash is given, ‘nano’
will read data from standard input.  This means you can pipe the output
of a command straight into a buffer, and then edit it.
```

File System Navigation: step 1

The Linux file system is organized in a hierarchical structure with the root directory (/) at the top.

The user will start in their home directory: /home/<myBamaID>

When you login to the server, enter the command pwd to see what directory you're currently in:

```
[myBamaID@uahpc-login001~] $pwd

/home/myBamaID
```

File System Navigation: step 2

Use the ls command to know what files are in the current directory.

```
[myBamaID@uahpc-login001 ~]$ ls
[myBamaID@uahpc-login001 ~]$
```

Since we have not created any files or directories it won't return anything.

File System Navigation: step 3

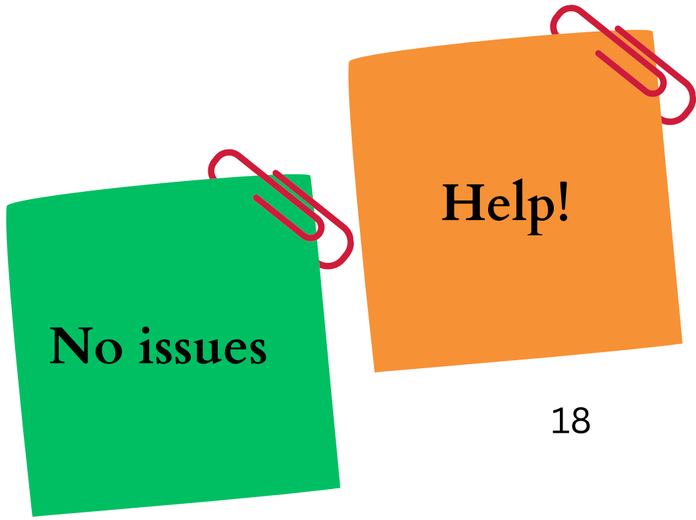
Use mkdir <directory name> to create a directory.

```
[myBamaID@uahpc-login001 ~]$ mkdir test
[myBamaID@uahpc-login001 ~]$
```

If it was successful, you will see the system prompt like in the example above.
Now, let's do ls again:

```
[myBamaID@uahpc-login001 ~]$ ls
test
```

We can now see the directory we just created.



File System Navigation: step 4

Use cd <directory-name> to go into the test directory we created and enter pwd before and after to see this.

```
[myBamaID@uahpc-login001 ~]$ pwd
/home/myBamaID
[myBamaID@uahpc-login001 ~]$ cd test
[myBamaID@uahpc-login001 test]$ pwd
/home/myBamaID/test
```

To go back one directory, use cd .. where .. represents the parent directory. Use pwd to see the changes take place.

```
[myBamaID@uahpc-login001 test]$ cd ..
[myBamaID@uahpc-login001 ~]$ pwd
/home/myBamaID
```

You can also use cd ~ to go back to your home directory from anywhere in the file system.
Let's go back into the test directory once more.


```
[myBamaID@uahpc-login001 ~]$ cd test
```

Nano: Text editor

To create/edit a file, we can use text editors such as nano, vim, or emacs. We will use nano as it is easy to use.

To edit a file using nano, type the following command: **`nano <filename>`**

If the file does not currently exist in the directory, nano will create a new one.

When you enter the command, you should see a window like this:

Nano: Text editor

The most important commands:

Ctrl + O : Save the file

Ctrl + X : Exit the editor

To learn more about nano, you can type man nano in the terminal to view the manual page for the nano command, or you can check out the nano documentation.

Creating a Python file

```
nano hpc_hello.py
```

in nano window:

```
#!/usr/bin/env python3
# hpc_hello.py: Print out a greeting

print('Hello World, from UAHPC!')
```

Ctrl + O # saves the file (after you hit Ctrl + O, hit enter again)
Ctrl + X # exits nano

To submit a job, we use a scheduler

On an HPC system, the scheduler manages which jobs run where and when.

Three **key functions of a scheduler**:

1. Allocates access to resources for users for some duration of time to perform work.
2. Provides a framework to start, execute, and monitor work.
3. Arbitrates contention for resources by managing queue of work.

Slurm

Any command or set of commands we want to run is a **job.**
The process of using a scheduler to run the job is called **batch job submission.**
We **need to monitor our jobs** and use of resources.

Job submission commands

sbatch – submit a batch script for later execution

salloc – obtain a job allocation

srun – obtain a job allocation and execute an application*

Accounting commands

sacct – display accounting data

sreport – report job usage and cluster utilization

Job management commands

squeue – view info about jobs

sinfo – view all info about nodes and partitions

scancel – signal jobs

To create a batch script for batch job submission

```
nano hello.batch
```

In nano window:

```
#!/bin/bash
#SBATCH --nodes=1
#SBATCH --mem=1G
#SBATCH --job-name=test_hpc_job
#SBATCH --partition main
#SBATCH --qos main
#SBATCH --time=1:00
#SBATCH --error errors.%A
#SBATCH --output output.%A

python ./hpc_hello.py
```

Ctrl + O # saves the file (after you hit Ctrl + O, hit enter again)
Ctrl + X # exits nano

Once you have exited nano,
run the Python file with:

```
sbatch hello.batch
```

You can check on the job with:

```
squeue -u myBamaID
```

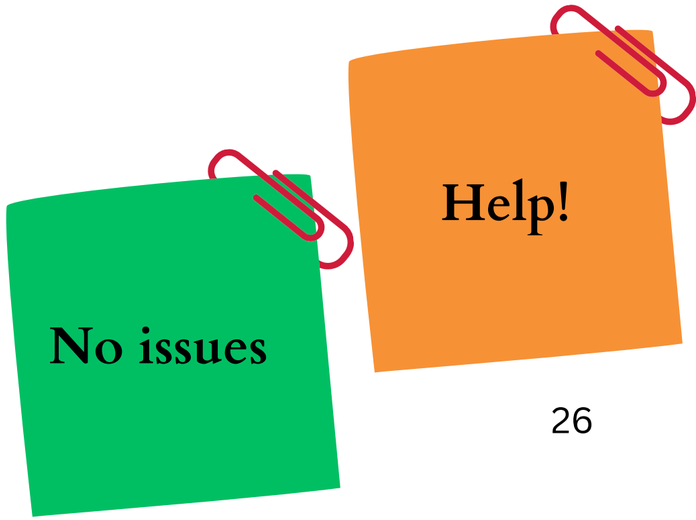
View the results of your first job on UAHPC!

To view a file, you can use: **`cat <filename>`**

Use this to view the output you just created:

```
[myBamaID@uahpc-login001 test]$ ls
[myBamaID@uahpc-login001 test]$ cat output.%A
```

`%A denotes job number`



SFTP: SSH File Transfer Protocol

A network protocol used for secure file transfer over a connection

Ensures data is transmitted encrypted

Differs from SCP (Secure Copy Protocol)

- Offers a range of commands to manage files and directories
- Establishes a persistent connection during file transfer, allowing resumption of interrupted transfers and multiple file management

Basic SFTP Commands

ls - Lists the contents of the current directory on the remote server. You can also ls to specify a path:

```
ls /path/to/remote/directory
```

lls - Lists the contents of the current directory on your local machine. You can also use lls to specify a path:

```
lls /path/to/local/directory
```

cd - Changes the current directory on the remote server

```
cd /path/to/remote/directory
```

lcd - Changes the current directory on your local machine

```
lcd /path/to/local/directory
```

pwd - Prints the current working directory on the remote server

```
pwd
```

lpwd - Prints the current working directory on your local machine

```
lpwd
```

mkdir - Creates a new directory on the remote server

```
mkdir new_remote_directory
```

lmkdir - Creates a new directory on your local machine

```
lmkdir new_local_directory
```


exit - To exit an SFTP session

SFTP: SSH File Transfer Protocol

In a new terminal, to connect to the SFTP server:

`sftp myBamaID@uahpc.ua.edu`

You will be prompted to enter your password and complete Duo two factor verification.

After successfully connecting to the server, you should see a system prompt like this: `sftp>`

SFTP: Copying files to and from a server

Local machine to remote server: put

```
put /path/to/localfile path/to/destination/folder
```

Remote server to local machine: get

```
get /path/to/remotefile path/to/local/directory
```

To copy a folder, use –r flag

SFTP Exercise

1. Navigate to Box folder: https://alabama.app.box.com/folder/383567545827
2. Download primes.batch and primesspiral.py
3. In new terminal, type the following and complete authentication steps:

```
sftp myBamaID@uahpc.ua.edu
```

4. SFTP the files, using these commands one by one:

```
put C:\Users\myBamaID\Downloads\spiral.batch /home/myBamaID/test
put C:\Users\myBamaID\Downloads\primesspiral.py /home/myBamaID/test
```

Don't close that terminal window!

Modules on HPC are a way to manage different software environments

Commands

module avail - lists all available modules

module load - load a module

module list - lists currently loaded modules

module purge - unloads all modules

Reasons we use modules

- allow users to load and unload software packages as needed
- helps in managing different versions of software and their dependencies

Using modules

BACK IN SSH TERMINAL

To load a module, use:

```
[myBamaID@uahpc-login001]$ module load python
[myBamaID@uahpc-login001]$ module list
Currently Loaded Modulefiles:
 1) slurm/uahpc/24.05.4   2) gcc/11.2.0   3) python/python3/3.11.8.test123
[myBamaID@uahpc-login001]$ module unload python

[myBamaID@uahpc-login001]$ cat primesspiral.py
```

```
import matplotlib.pyplot as plt
import numpy as np

# Check if a number is prime
def is_prime(n):
    if n < 2:
        return False
    for i in range(2, int(n**0.5)+1):
        if n % i == 0:
            return False
    return True

# Generate spiral coordinates
def generate_spiral(n_points, spacing=0.5):
    theta = np.linspace(0, n_points * spacing, n_points)
    r = theta
    x = r * np.cos(theta)
    y = r * np.sin(theta)
    return x, y

# Number of points to plot
N = 1000
# Generate spiral
x, y = generate_spiral(N)

# Plot
plt.figure(figsize=(8, 8))
for i in range(N):
  if is_prime(i):
    plt.plot(x[i], y[i], 'ro', markersize=3) # red dot for prime
  else:
    plt.plot(x[i], y[i], 'k.', markersize=1) # small black dot for non-prime
```

Required packages are at top

Where numpy is used

Where matplotlib is used

How do we get the packages we need?

We use a package manager.

A package manager is a system that enables you to set up a **new, project specific software environment** containing specific software versions as well as the versions of **additional packages and required dependencies** that are **all mutually compatible.**

Why use a package manager?

- They help resolve dependency issues by allowing you to use different versions of a package for different projects.
- Your projects are now self-contained and reproducible by capturing all package dependencies in a single requirements file.
- Allows you to install packages on a host on which you do not have admin privilege.

Setting up our Environment

Run this code line by line:

```
cd .. # should be out of test directory
module load miniconda3/base

conda create --name day1
conda activate day1


conda install -c conda-forge python numpy matplotlib -y

#Once install is complete
conda deactivate
```

Running a job using the new environment

Submit the job

`sbatch spiral.batch`

Monitoring job status

`squeue -u myBamaID`

View the output

`cat output_primes.%A`

BACK IN SFTP WINDOW

Retrieve the .png file

`get /home/myBamaID/test/spiral.png C:\Users\myBamaID\Desktop`

```
#!/bin/bash
#SBATCH --job-name=spiraljob
#SBATCH --output=output_primes.%A
#SBATCH --error=error_primes.%A
#SBATCH --partition=main
#SBATCH --qos=main
#SBATCH --nodes=1
#SBATCH --mem=2G
#SBATCH --time=00:10:00

module purge
module load miniconda3/base
conda activate day1

python ./primesspiral.py
```

Loading the module

Activating the conda environment

Final checks

Check on your account

```
[myBamaID@uahpc-login001 ~]$ sacct -u myBamaID
```

View info on nodes and partitions

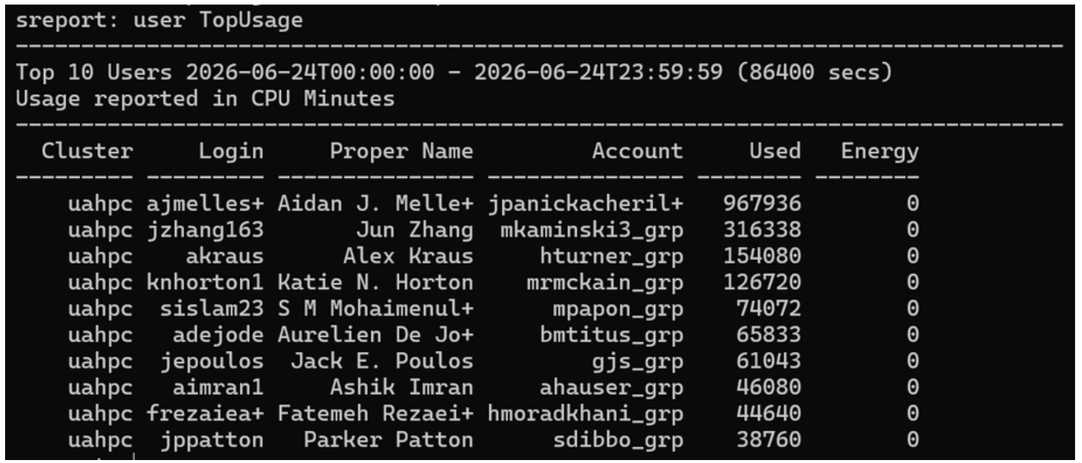
```
[myBamaID@uahpc-login001 ~]$ sinfo
```

Return a report about top users

```
[myBamaID@uahpc-login001 ~]$ sreport
sreport: user TopUsage
```

Exit the ssh session

```
[myBamaID@uahpc-login001 ~]$ exit
```



```
sreport: user TopUsage
--------------------------------------------------------------------------------
Top 10 Users 2026-06-24T00:00:00 - 2026-06-24T23:59:59 (86400 secs)
Usage reported in CPU Minutes
--------------------------------------------------------------------------------
  Cluster     Login     Proper Name         Account     Used   Energy
--------- --------- --------------- --------------- -------- --------
    uahpc ajmelles+ Aidan J. Melle+ jpanickacheril+   967936        0
    uahpc jzhang163       Jun Zhang  mkaminski3_grp   316338        0
    uahpc    akraus      Alex Kraus     hturner_grp   154080        0
    uahpc knhorton1 Katie N. Horton    mrmckain_grp   126720        0
    uahpc  sislam23 S M Mohaimenul+      mpapon_grp    74072        0
    uahpc   adejode Aurelien De Jo+     bmtitus_grp    65833        0
    uahpc  jepoulos  Jack E. Poulos         gjs_grp    61043        0
    uahpc   aimran1     Ashik Imran     ahauser_grp    46080        0
    uahpc frezaiea+ Fatemeh Rezaei+ hmoradkhani_grp    44640        0
    uahpc  jppatton   Parker Patton      sdibbo_grp    38760        0
```

LUNCH BREAK

Please return by 1:30pm

Qualtrics survey